DIPTIMAN MOHANTA

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Education

Parala Maharaja Engineering College, Berhampur **December 2021 – June 2025**
Bachelor of Technology – Electrical Engineering | CGPA: 7.89/10 *Berhampur, Odisha, India*

St. Xavier's High School, Keonjhar **August 2019 – July 2021**
Intermediate in Science | Percentage: 79.8% *Keonjhar, Odisha, India*

Vikash Convent School, Karanjia **2012 – 2019**
Matriculation | Percentage: 86.6% *Karanjia, Odisha, India*

Research & Internship Experience

Project Associate – Automatic Speech Processing **September 2025 – Ongoing**
SPIRE Lab, Indian Institute of Science (IISc), Bengaluru

- Conducting research in automatic speech processing for low-resource Indian languages within the Signal Processing, Interpretation and REpresentation (SPIRE) Laboratory.

Research Intern – Complex-Valued Neural Networks for ECG Classification **January 2025 – June 2025**
Perception & Intelligence Lab (PIL), IIT Kanpur

- Developed a Complex-Valued Neural Network (CVNN) for Atrial Fibrillation (AF) classification using Gramian Angular Field (GAF) image representations of ECG signals.
- Architected a hybrid CNN combining real and complex-valued layers to model phase-magnitude dynamics in GASF/GADF representations.

Research Intern – Contactless Vital Sign Detection **June 2024 – August 2024**
Defence Institute of Advanced Technology (DIAT), DRDO, Pune

- Designed and developed a Continuous Wave (CW) radar system for contactless detection of vital signs including heart rate and respiratory rate.
- Implemented advanced signal processing pipelines for data acquisition and feature extraction from raw radar returns.

Embedded Systems Intern **April 2023 – May 2023**
Emertxe – Institute for Embedded Systems and IoT

- Completed training in Embedded C programming, microcontroller utilization, and sensor integration; delivered a functional home automation system as a capstone project.

Research and Technical Projects

RUL Prediction of PV Systems Under Dynamic Environmental Conditions **August 2024 – December 2024**

- Developed a semi-parametric prognostic framework for solar PV Remaining Useful Life (RUL) forecasting, combining cumulative damage modeling with multivariate Bernstein bases.
- Achieved superior prediction accuracy over baseline methods when validated on real PV operational data, improving reliability and cost-efficiency of renewable energy systems.

Micro-Doppler Signature-Based Target Classification Using FMCW Radar **August 2024 – November 2024**

- Built a target classification system using FMCW radar micro-Doppler signatures to distinguish between drones and birds.
- Applied spectrogram analysis and deep learning methods, significantly enhancing classification accuracy over traditional signal processing baselines.

EEG Lie Detection Using Wavelet-Based Feature Extraction and Ensemble Methods **July 2024 – February 2025**

- Developed a lie detection system using EEG signals, extracting wavelet-domain features to differentiate truthful from deceptive responses.
- Implemented ensemble classifiers to achieve robust identification of deception from brainwave patterns.

ISRO Robotics Challenge – URSC 2024 **November 2023 – May 2024**

- Contributed to developing an autonomous rover; specialized in PCB design for microprocessors, driver modules, and sensor integration.
- Led manipulator control and positioning using MATLAB, enabling autonomous navigation, object identification, and size estimation.

Automatic Guided Rocket Fin Stabilizer Mechanism **September 2023 – November 2023**

- Designed an automatic guided fin stabilizer mechanism for rockets, ensuring precise trajectory control independent of external disturbances and enhancing navigation reliability.

Drone-Based Crop and Vegetation Monitoring System **July 2023 – September 2023**

- Led development of a specialized agricultural drone for vegetation assessment using NDVI calculations.
- Implemented algorithms to predict water and fertilizer requirements and enable early detection of plant stress, diseases, and nutrient deficiencies.

Open Source Contributions

sigclean | *Python* | github.com/diptiman-mohanta/sigclean **2025**

- Developed a comprehensive Python library for cleaning and preprocessing biomedical signals (ECG, EEG, EMG). Provides a complete filtering, denoising, and segmentation toolkit for physiological signal pipelines.

SigFeatX | *Python* | github.com/diptiman-mohanta/SigFeatX **2025**

- Built a Python library for extracting statistical features from 1D signals using advanced decomposition techniques and signal processing metrics.

Publications

Journals

[1] P. S. P. K. Mahali, D. Mohanta, and T. Sandhan, "Complex afnet: A hybrid complex-valued deep network for atrial fibrillation detection," *IEEE Transactions on Artificial Intelligence*, pp. 1–12, 2026. DOI: 10.1109/TAI.2026.3671218.

Magazines

[1] D. Mohanta and S. K. Sahoo, "Delving into the world of cables and wires – part 1 & part 2," *Electrical India Magazine*, Aug. 2024.

[2] D. Mohanta and S. K. Sahoo, "Environmental impacts of solar panels," *Electrical India Magazine*, Mar. 2024.

Achievements

SPRINT East Edition, iHUB-AWaDH, IIT Ropar – Winner of Strategic Program for Research Innovation and Next-Gen Tech Commercialization (SPRINT) East Edition.

ISRO Robotics Challenge – URSC 2024 – Qualified the Preliminary Round of the ISRO Robotics Challenge.

Technical Skills

- **Programming Languages:** C, Python
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, MATLAB Deep Learning Toolbox
- **Tools:** MATLAB, Simulink, ROS, EasyEDA, Proteus 8 Professional
- **Hardware:** Raspberry Pi, STM32, ESP-32, ESP-8266, Arduino, Picoscope
- **Operating Systems:** Linux, Windows

Research Interests

Automatic Speech Recognition, Deep Learning, Signal Processing, Biomedical Signal Analysis, Radar Systems, Image Processing, Internet of Things, Embedded Systems

Extracurricular Activities

- **Technical:** Active member of college Robotics Club; participant in national-level design competitions.
- **Sports & Hobbies:** Kho-Kho, Swimming, Photography.